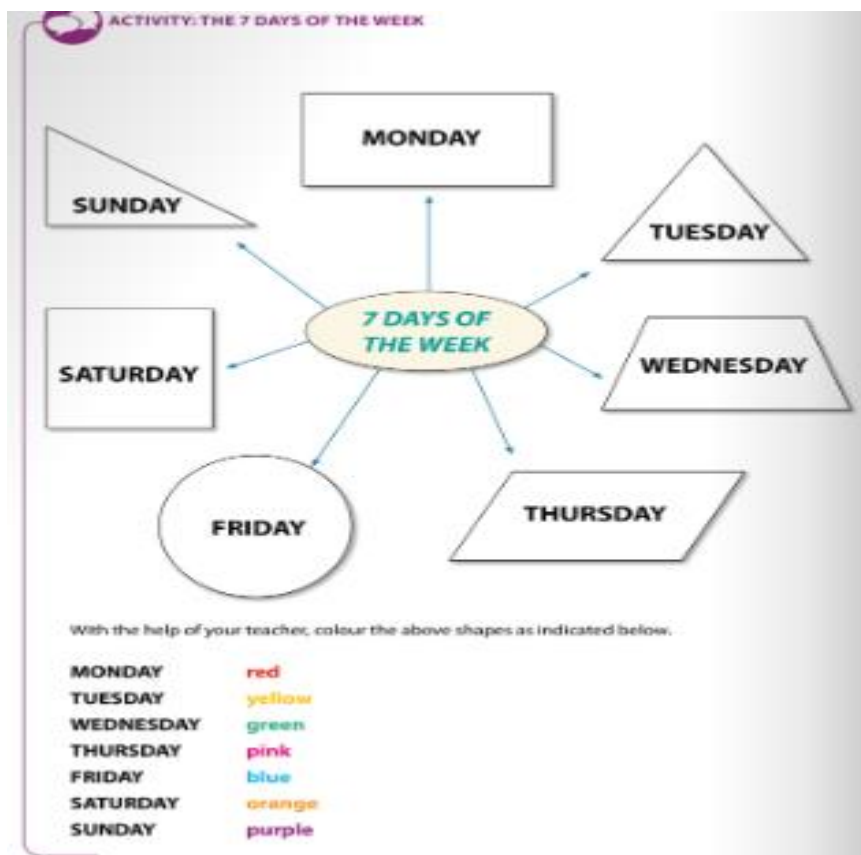


Grade	Activity
Grade 7 Theme: Welcome to school	- Days of the week (pages 47-48; page 155) - Months of the year (pages 48-49; pages 156-157) - Construct a clockface and work with time (pages 50-51, page 53; pages 139-142) Book: Foundation Mathematics (Extended Programme) Grade 7 (Teacher's Book Part 1)
Grade 8 Theme: Road safety	- Back to school (shapes of road signs) (pages 1-3) Book: Let's Learn Mathematics Grade 2 (Teacher's Book Part 1) - Match objects with shapes (page 2) Book: Let's Learn Mathematics Grade 3 (Teacher's Book Part 1) Remarks: The educator needs to initiate a discussion on road accidents and possible causes. This should be followed by the importance of road safety and understanding different road signs (together with their corresponding shapes).
Grade 9: Les Métiers	- Measurement of length (pages 57-60) Book: Let's Learn Mathematics Grade 4 (Teacher's Book Part 2) - Measure length (m and cm) in real-life situations (Pages 72-73) - Going around the classroom (pages 183-184) Book: Let's Learn Mathematics Grade 5 (Teacher's Book Part 2) - Perimeter (Activity 13: 2 pages)

	Book: Fortified Learning Environment (FLE) Toolkit: Mathematics Functional Numeracy Designing a photo frame (Unit 18: 2 pages) Book: Fortified Learning Environment (FLE) Toolkit: Mathematics Functional Numeracy
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Grade 7 Resources

Days of the Week





EXERCISE

Try these.



1. Write the days of the week in the correct order.

Wednesday - Saturday - ~~Monday~~ - Friday - Tuesday - Sunday - Thursday

1. **Monday**
2.
3.
4.
5.
6.
7.



EXERCISE

Recapitulate the days of the week.
Try these.



1.

1 st day	Wednesday
2 nd day	Monday
3 rd day	Friday
4 th day	Tuesday
5 th day	Thursday

2. Complete the following.

- (a) Monday, Tuesday, _____, Thursday.
- (b) Friday, _____, Sunday.
- (c) Wednesday, _____, _____, Saturday.

Months of the Year

Calendar 2018

January 2018	February 2018	March 2018	April 2018
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9
10	10	10	10
11	11	11	11
12	12	12	12
13	13	13	13
14	14	14	14
15	15	15	15
16	16	16	16
17	17	17	17
18	18	18	18
19	19	19	19
20	20	20	20
21	21	21	21
22	22	22	22
23	23	23	23
24	24	24	24
25	25	25	25
26	26	26	26
27	27	27	27
28	28	28	28
29	29	29	29
30	30	30	30
31	31	31	31

May 2018	June 2018	July 2018	August 2018
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9
10	10	10	10
11	11	11	11
12	12	12	12
13	13	13	13
14	14	14	14
15	15	15	15
16	16	16	16
17	17	17	17
18	18	18	18
19	19	19	19
20	20	20	20
21	21	21	21
22	22	22	22
23	23	23	23
24	24	24	24
25	25	25	25
26	26	26	26
27	27	27	27
28	28	28	28
29	29	29	29
30	30	30	30
31	31	31	31

September 2018	October 2018	November 2018	December 2018
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9
10	10	10	10
11	11	11	11
12	12	12	12
13	13	13	13
14	14	14	14
15	15	15	15
16	16	16	16
17	17	17	17
18	18	18	18
19	19	19	19
20	20	20	20
21	21	21	21
22	22	22	22
23	23	23	23
24	24	24	24
25	25	25	25
26	26	26	26
27	27	27	27
28	28	28	28
29	29	29	29
30	30	30	30
31	31	31	31

1 Jan. New Year's Day
 2 Jan. New Year's Holiday
 3 Jan. New Year's Holiday
 4 Jan. New Year's Holiday
 5 Jan. New Year's Holiday
 6 Jan. New Year's Holiday
 7 Jan. New Year's Holiday
 8 Jan. New Year's Holiday
 9 Jan. New Year's Holiday
 10 Jan. New Year's Holiday
 11 Jan. New Year's Holiday
 12 Jan. New Year's Holiday
 13 Jan. New Year's Holiday
 14 Jan. New Year's Holiday
 15 Jan. New Year's Holiday
 16 Jan. New Year's Holiday
 17 Jan. New Year's Holiday
 18 Jan. New Year's Holiday
 19 Jan. New Year's Holiday
 20 Jan. New Year's Holiday
 21 Jan. New Year's Holiday
 22 Jan. New Year's Holiday
 23 Jan. New Year's Holiday
 24 Jan. New Year's Holiday
 25 Jan. New Year's Holiday
 26 Jan. New Year's Holiday
 27 Jan. New Year's Holiday
 28 Jan. New Year's Holiday
 29 Jan. New Year's Holiday
 30 Jan. New Year's Holiday
 31 Jan. New Year's Holiday



With the help of your teacher, discuss the following with your friends.

- How many months are there in a year?
- Ask your friend to point out and read the name of the months.
- How many public holidays are there in December?
- On which date is the public holiday in December and on which occasion?
- How many public holidays are there in February?
- How many Sundays are there in July?
- How many days are there in February?

There are 12 months in a year.



EXERCISE



Complete the following calendar.

1.

August 2018						
Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
					3	
	6					
			15			
				30		

- How many days are there in this month?
- How many Sundays were there in August 2018?
- What day is the 24th of August 2018?



Complete the following.

- List the name of months starting with the letter **J**.

 - List the name of months starting with the letter **M**.



Use the given words below to complete the following.
December, March, February, June

- The second month of the year is _____.
 - The last month of the year is _____.
 - The sixth (6th) month is _____.
 - The month before the 4th month is _____.

Constructing a clockface and reading time

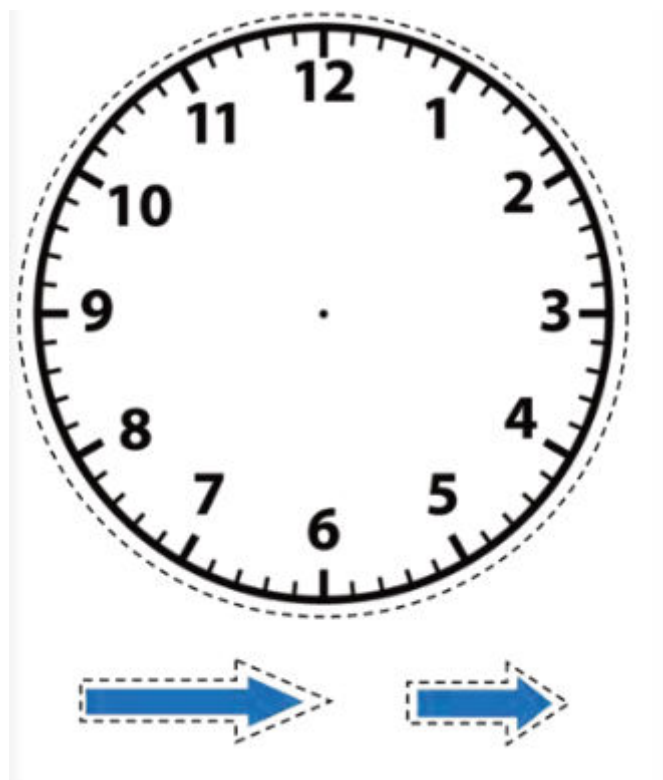


ACTIVITY: CREATING A CLOCK FACE AND READING TIME

Materials needed: a paper plate, glue, a cork and a pin, clock face template, hour hand, minute hand.

With the help of your teacher:

- cut out the template on the following page and glue it on the paper plate.
- then cut out the hour hand and the minute hand.
- make a small hole in the centre of the clock face.
- place the hands in the middle of the clock face.
- fix the 2 hands, the longer on top, to the centre of the clock face, with a pin.
- use the cork to cover the pointed end of the pin.



- Listen carefully to the instructions from your teacher to read time using o'clock.
- When the long hand is pointing to 12, the time is read by looking at the minute hand and reading the number to which it is pointing.



EXERCISE



Try these.

1. Draw the hour hand in the correct position.

(a)



4 o'clock

(b)



8 o'clock

2. Write the correct time.

(a)



(b)





ACTIVITY - USING CLOCK FACE TO SHOW TIME

- With the help of your teacher use your clock face to read time using "o'clock".
- When the long hand is pointing to 12, the time is read by looking at the minute hand and reading the figure to which it is pointing.
- Example: If the long hand points to 12, and the short hand points to 9, then the time is read as **9 o'clock**.



- **Be careful not to turn any hand anticlockwise.**
- Turn it 360° clockwise as many times as required but never in the anticlockwise direction.



CHECK THIS LINK

The link below has a fun and interactive game on the months of the year. This will help the students to answer questions correctly and also cater for awareness of ordinal numbers.
http://www.abcy.com/months_of_the_year.htm
http://www.abcy.com/days_of_the_week.htm



EXERCISE



Try these.

1. Use your clock face and work in pairs or in groups to show the time

- | | |
|-------------------|--------------------|
| (a) Three o'clock | (b) Eleven o'clock |
| (c) Five o'clock | (d) Eight o'clock |

2. Match the following.



Nine o'clock



Three o'clock



Ten o'clock

3. Write down the times shown on each clock face.



In figures :

In words :



In figures :

In words :

ACTIVITY - WORKING WITH HALF PAST

Use your clock face or construct one if you do not have it.



- Draw a vertical line from 12 to 6.
- Fold the clock face along the line and observe that the clock face has been divided into two halves.
- With the help of your teacher or your friends put the hour hand exactly on 2.
- Then put the minute hand exactly on 6.
- This time is read as half past the hour (here it is half past two).
- On a real clock, the hour hand will be exactly halfway between 2 and 3.

EXERCISE



Match the following.



Half past seven



Half past three



Half past nine



ACTIVITY - WORKING WITH QUARTER PAST THE HOUR

- Use your clock face in which you have already drawn the vertical line.
- Now draw a horizontal dotted line from 9 to 3.
- Then fold the clock face along the horizontal line.
- Observe that your clock face has been divided into 4 equal parts.
- The time is read as "quarter past 2".
- On a real clock, the hour hand will have moved a little just after 2.



With the help of your teacher, show the following times on your clock face. You may work in pairs.

- Quarter past four
- Quarter past nine
- Quarter past six



EXERCISE



Complete each clock face.



Half past one



Half past eleven



Quarter past eight



Quarter past seven

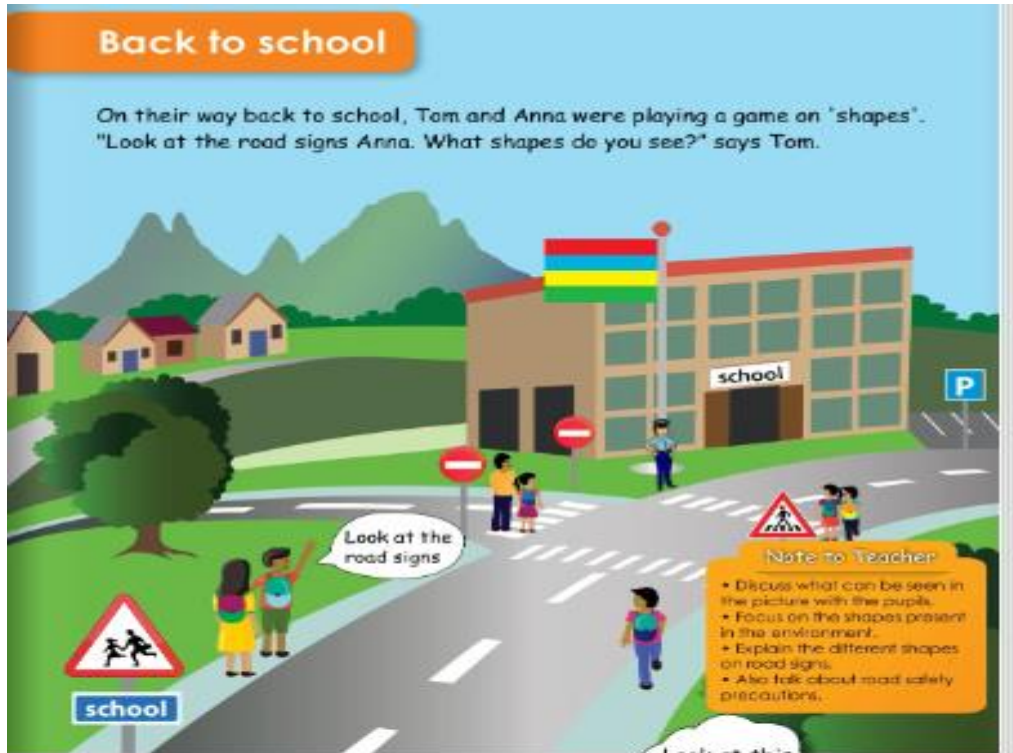
Grade 8

Grade 8 Resources

Road Safety

Back to school

On their way back to school, Tom and Anna were playing a game on 'shapes'.
"Look at the road signs Anna. What shapes do you see?" says Tom.



Look at the road signs

Look at this picture.

Note to Teacher

- Discuss what can be seen in the picture with the pupils.
- Focus on the shapes present in the environment.
- Explain the different shapes on road signs.
- Also talk about road safety precautions.

- 1 Observe carefully and discuss the picture.
- 2 What are the different shapes that you can see in the picture?

1



Do you remember?



circle



rectangle



triangle



square

Note to Teacher

Build upon the start story to create a hands-on activity.

Game: Shapes in the bag

Materials:

Shapes made from carton cards/foam paper
An opaque bag

- A pupil is asked to put his/her hand in a bag to choose one of the shapes inside. By exploring the shape with his/her hand while it is still in the bag, he/she guesses what shape it is.
- Let the pupil take out the shape he/she has chosen and let him/her describe it. (Use various objects of the same shape)

You can ask the following questions:

Can you find other objects in the bag that have the same shape?

Let's compare some objects. How many objects have the same shape?

Activity: Road Signs



1 Look at these road signs.

2 With the help of your teacher, identify and name more shapes that you can find in the class.

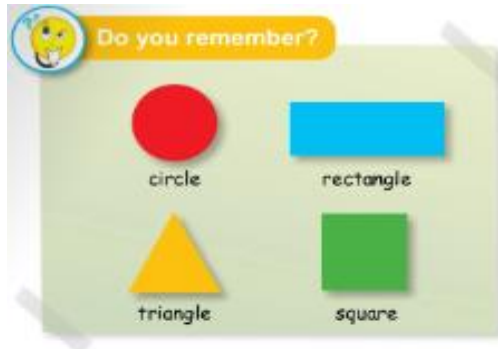
Let's work out

1. I match the shapes with the road signs.

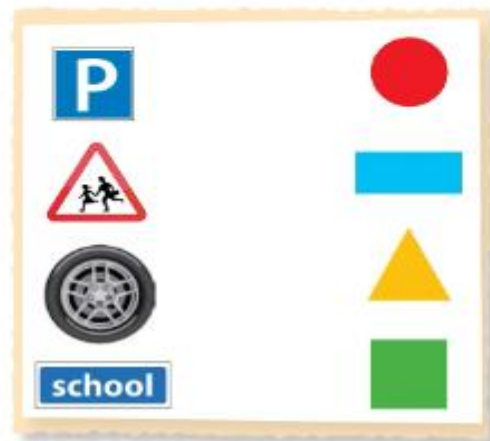


Match each shape to the correct picture.





Activity: I match the objects with the shapes



Note to Teacher

It is good to start the chapter with an activity done in Grade 2 such as:

Game: Shapes in the bag. To guess the shapes in the bag.

Questioning (oral):

- What can you say about the rectangle?
- Do the rectangle and the square look alike?
- What are the differences?
- List the differences between a triangle and a rectangle.
- What can you say about a circle and other shapes?

Some scaffolding activities:

Materials:

- Flash cards
- Description of objects
- Picture to picture matching
- Picture to word matching
- Word to word matching
- Jumble letters

Note:

It is good to teach the pupils to read the instructions for the activities.

Grade 9

Grade 9 Resources

Les Métiers

Measurement: Length and Perimeter

Length

During a class, the teacher measures the length of the whiteboard using his hand span. He asks one pupil to do the same.



Teacher



Pupil

1 Observe carefully and discuss the pictures.

2 Do you think the teacher and the pupil will get the same number of hand spans? Why?

Activity 1

Measuring length using an arbitrary unit

Carry out the above activity and record the results in the table below.

	Number of hand spans
Teacher	
Pupil 1	
Pupil 2	
Pupil 3	
Pupil 4	

After the activity, let the pupils deduce that it is not convenient to use hand spans to measure length as the number of hand spans may differ.

Are the number of hand spans equal for all (teacher and pupils)?

What can you conclude?

12



Did you know?

A draper uses a ruler to measure cloth.



Note to Teacher

- Bring a metre rule in class so that the pupils can observe and use it to measure length.

Note: A draper sells cloth.

Observe the picture carefully. Is the draper using the same ruler as yours?

The metre

The draper uses a special ruler called a **metre rule**.

Note: The metre rule shown below is not to scale.

The **metre** is the **standard unit** of length. The symbol for metre is **m**.



The **metre** is used to measure the length or height of objects.

Note to Teacher

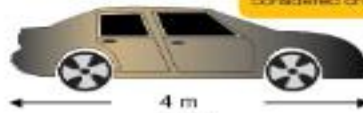
- Lay emphasis on the metre rule.
- The unit cm will be considered on page 40.

3m



a wall

The wall is 3 metres (3 m) high.



4 m

a car

The car is 4 metres long.

Activity 2 The metre

With the help of your teacher, estimate whether the following lengths are less than or greater than one metre.

Objects	Less than / greater than
Pencil 	
Blackboard 	
Book 	
Whale 	
Bus 	

Working with length

We can add, subtract, multiply and divide the lengths of objects.

Example: Work out the following.

(a) $9\text{ m} + 3\text{ m}$

$$\begin{array}{r} 9\text{ m} \\ + 3\text{ m} \\ \hline 12\text{ m} \end{array}$$

(b) $25\text{ m} - 12\text{ m}$

$$\begin{array}{r} 25\text{ m} \\ - 12\text{ m} \\ \hline 13\text{ m} \end{array}$$

(c) $14\text{ m} \times 2$

$$\begin{array}{r} 14\text{ m} \\ \times 2 \\ \hline 28\text{ m} \end{array}$$

(d) $45\text{ m} \div 5$

$$\begin{array}{r} 5 \overline{) 45\text{ m}} \\ \underline{09\text{ m}} \\ 0 \end{array} = 9\text{ m}$$

Try These

1. Work out the following.

(a) $2\text{ m} + 5\text{ m} = \underline{\hspace{2cm}}\text{ m}$

(b) $7\text{ m} + 8\text{ m} = \underline{\hspace{2cm}}\text{ m}$

(c) $12\text{ m} - 3\text{ m} = \underline{\hspace{2cm}}\text{ m}$

(d) $9\text{ m} - 2\text{ m} = \underline{\hspace{2cm}}\text{ m}$

(e) $3 \times 2\text{ m} = \underline{\hspace{2cm}}\text{ m}$

(f) $5 \times 4\text{ m} = \underline{\hspace{2cm}}\text{ m}$

(g) $15\text{ m} \div 3 = \underline{\hspace{2cm}}\text{ m}$

(h) $30\text{ m} \div 6 = \underline{\hspace{2cm}}\text{ m}$

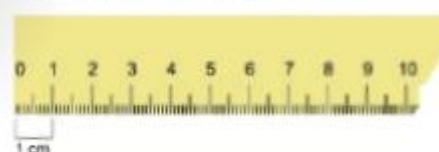
Activity 3 The centimetre, cm

To measure the length of small objects, a smaller unit is used.

A smaller unit is the **centimetre**.

The symbol for centimetre is **cm**.

The ruler used in class has the unit **cm**.



Note to Teacher

Materials needed: 30 cm ruler or shorter.
Some objects are less than 30 cm in length, e.g. copybook, books, pencil case.

Now, use your ruler to measure and record the lengths of some objects in the table below.

Object	Length in cm
Mathematics book	
Copybook	
Pencil	
Pencil case	
Eraser	

Try These

2. Use your ruler to draw lines of the following lengths.

- (a) 3 cm
- (b) 8 cm
- (c) 5 cm

NOTE TO TEACHER

Activity: Measure length (m/cm) in real-life situations

Activity needs to be done with the whole class.

Materials needed: a measuring tape, ruler, a 1 m ribbon, a 1 m rope.

Procedure:

- Show the objects to the students.
- Let them deduce that the length of the ribbon and rope are of the same length.
- Let the students measure different objects of different lengths in the classroom and outside. Show them how to measure correctly using the measuring tape and ruler.
- Let the students estimate the length around their waists.
- Using a measuring tape, ask them to measure the length around their waists.
- What do they notice about the measurements?

Activity: Problems involving length

Activity to be done with the whole class.

- Show the students that length can be added.
- Let them find a situation where they can add length.
- Ensure that the students are at ease with the four operations.
- Let the students carry out the exercises.

Activity: Compare and convert metre and centimetre

Materials needed: measuring tapes

Procedure:

- Put the students in pairs and give each pair a measuring tape.
 - Ask them to measure one metre.
 - Beneath the 1 m, they will find another measurement in cm.
 - Show them that 1 m = 100 cm.
 - Ask them to find the number of cm in
- (a) 2 m (b) 5 m (c) 10 m



ACTIVITY 3: MEASURE LENGTH (m/cm) IN REAL-LIFE SITUATIONS

Materials needed: a measuring tape, ruler, a 1 m ribbon, a 1 m rope.



Your teacher will show you the different objects. Follow carefully his/her instructions. You will need a measuring tape and a ruler.

- Measure the length of the ribbon and the rope.
- What do you notice?
- Measure different objects in the classroom and outside.
- Estimate the length around your waist.
- Using a measuring tape, measure the length around your waist.
- What do you notice about the measurements?

NOTE TO TEACHER

Activity : Going Around the classroom.

Materials needed: measuring tapes, pencil and a sheet of paper

Procedure:

- Divide the students in 4 groups.
- Give each group a measuring tape, a sheet of paper and a pencil.
- Assign each group a side of the classroom to measure and note it down.
- When this is done, rotate the groups so that they measure the other sides.
- At the end, each group needs to measure the four sides.
- Discuss the measurements obtained.
 - Are the measurements the same?
 - What units have they been using to measure? Metres or centimetres?
 - Any difficulties encountered?
- Ask them what they will do if they need to find the distance all around the room in metres and centimetres.
- Help them understand that they need to add the four lengths obtained to find the distance around the room.
- Provide guidance where needed.

GOING AROUND THE CLASSROOM



1. Listen carefully to the instructions given by your teacher. You will work in groups and will need a measuring tape, a sheet of paper and a pencil.
2. Measure the side of the room assigned to you.
3. Note it down on the sheet of paper.
4. When you are done, measure the other sides of the room and note them down.
5. Discuss the measurements obtained with your friends.
 - (a) Are the measurements the same?
 - (b) What units have you been using to measure? Metres or centimetres?
 - (c) What difficulties have you encountered?
6. How will you find the distance around the room in **metres** and in **centimetres**?

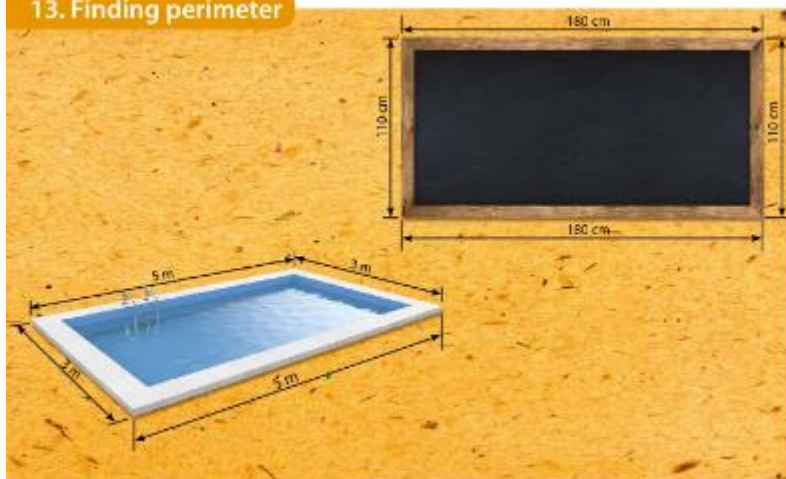


DID YOU KNOW

When you "walk-around" a figure to find the length or the distance, it means that you are finding the "**Perimeter**".

Perimeter comes from the Greek word "perimetros". Peri means "around" and metros means "measure".

13. Finding perimeter



This activity enables students to apply their knowledge of measurement to find perimeter.

13. Finding perimeter

Learning Outcomes:

- Read and measure using standard units of length (m, cm and mm)
- Find perimeter of given configuration

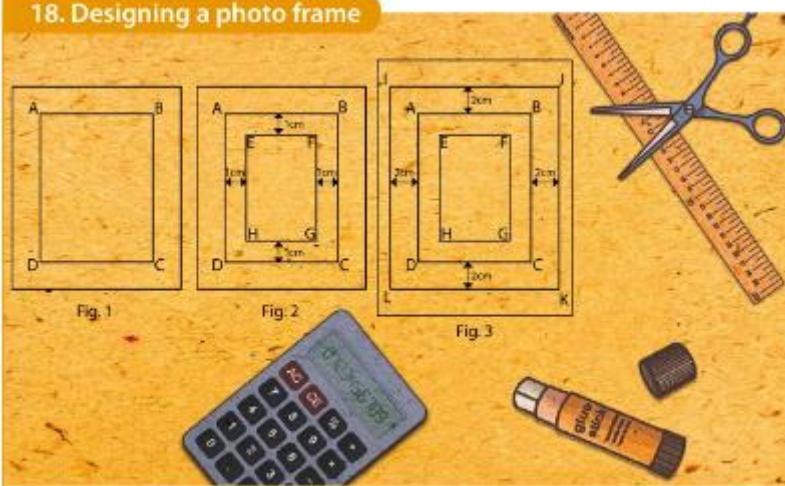
Materials needed: 30 cm ruler, measuring tape

Procedure:

1. Teacher encourages students to interpret the scale on a ruler and a measuring tape.
2. Students measure the dimensions of the classroom and the objects shown in the table below (in appropriate units).
3. Teacher discusses the importance of perimeter.
4. Students find the perimeter of the classroom and objects in the table below.

	Length	Width	Perimeter
Classroom dimension			
Teacher's table			
Student's desk			
Cupboard			
Whiteboard			

18. Designing a photo frame



This activity enables students to make sense of length (cm, mm) and perform simple calculations involving lengths.

18. Designing a photo frame

Learning Outcomes:

- Measure length (in cm, mm) using a ruler
- Draw rectangles of given dimensions

Materials needed: 30-cm ruler, calculator, photo (e.g., 15 cm x 10 cm), two pieces of bristol paper of A4 size (frame and base), scissors, glue and compass box

Procedure:

1. Ask students to measure the photo that is provided to them using their ruler (e.g. photo dimensions: 15 cm x 10 cm)
2. Ask students to place the photo, approximately in the middle of the A4 paper and trace the boundary. Encourage students to place the photo with the sides parallel to the sides of the A4 paper. Label this rectangle ABCD as in Fig. 1.
3. Measure 1 cm inside each of the 4 sides of the rectangle "ABCD" and draw another rectangle "EFGH" as in Fig. 2 (Encourage students to use their set square to construct right angles).
4. Now measure 2 cm outside each of the 4 sides of rectangle "ABCD". Draw and label the rectangle "IJKL" as in Fig. 3.
5. Cut out rectangle "IJKL".
6. Cut out rectangle "EFGH" by folding the paper into two equal parts.
7. You may use a cardboard for the base of the photoframe (same as "IJKL"). Stick the photo and the frame on the base. Students may decorate the border and design a stand to hold the frame.
8. Now, ask students to calculate the perimeter and area of the different rectangles and to record these in their copybook.

